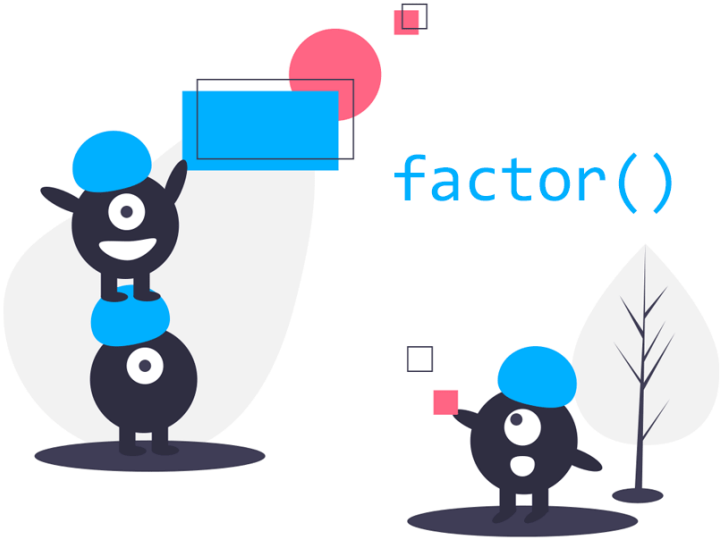




INSP/ESPM
Escuela de Salud Pública de México

R {lubridate}

José Luis Texcalac Sangrador

Procesamiento, visualización y modelado de datos espaciales en R

Importamos la malla de datos y generamos el resumen de los datos

```
contam <-
  read_csv("./data/promedios_2021_ps.csv", skip = 8) %>%
  clean_names() %>%
  print()
```

```
summary(contam)
```

```
       date          id_station      id_parameter      value             unit
 Length   :6692   Length   :6692   Length   :6692   Min.   :  2.00   Min.   :2
 N.unique : 243   N.unique :  28   N.unique :   2   1st Qu.: 17.00   1st Qu.:2
 N.blank  :   0   N.blank  :   0   N.blank  :   0   Median : 25.00   Median :2
 Min.nchar:  10   Min.nchar:   3   Min.nchar:   4   Mean   : 30.46   Mean   :2
 Max.nchar:  10   Max.nchar:   3   Max.nchar:   5   3rd Qu.: 39.00   3rd Qu.:2
                                                    Max.   :158.00   Max.   :2
```

- Transformamos variables a factor
- Renombramos columnas
- Seleccionamos columnas de interes
- Visualizamos el resumen de la información
- ¿qué variable nos falta editar?

```r
contam <-
  contam %>%
  mutate(
    site = factor(id_station),
    pollutant = factor(id_parameter)
  ) %>%
  rename(concentration = value) %>%
  select(-c(id_station, id_parameter, unit)) %>%
  print()
```

```r
summary(contam)
```

```
      date          concentration          site          pollutant
 Length   :6692   Min.   :  2.00   BJU     : 482   PM10 :3529
 N.unique : 243   1st Qu.: 17.00   MER     : 478   PM2.5:3163
 N.blank  :   0   Median : 25.00   PED     : 470
 Min.nchar:  10   Mean   : 30.46   TLA     : 432
 Max.nchar:  10   3rd Qu.: 39.00   SFE     : 386
                  Max.   :158.00   CAM     : 364
                                   (Other):4080
```

- Primero revisar el formato en el que está capturada la fecha, por ejemplo:
  - dd/mm/yy, dd/mm/yyyy
  - mm/dd/yy, mm/dd/yyyy
- Una fecha formateada en R siempre tendrá este formato “yyyy-mm-dd”
  - 2021-01-01
  - Así la visualiza el software, después es posible editar la salida

```
> contam
# A tibble: 6,692 × 4
   date       concentration site  pollutant
   <chr>              <dbl> <fct> <fct>
 1 01/01/2021            70 ACO   PM10
 2 01/01/2021            44 ATI   PM10
 3 01/01/2021            73 CUT   PM10
 4 01/01/2021            43 FAC   PM10
 5 01/01/2021            56 MER   PM10
 6 01/01/2021            40 MER   PM2.5
 7 01/01/2021            29 NEZ   PM2.5
 8 01/01/2021            34 PED   PM10
 9 01/01/2021            25 PED   PM2.5
10 01/01/2021            59 SAG   PM10
# ℹ 6,682 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Fechas con código base

Columna o vector con fechas en algún formato consistente
- mes-día-año
- año/mes/dia

Formato de la fecha

```
as.Date(fecha,  format)
```

malla

fecha	valor
12/11/19	12.1
31/01/21	13.4
28/02/17	11.8
03/09/18	14.4
22/11/20	12.7

%d/%m/%y

¡Importante!
El formato se refiere a como está capturado en la columna original, NO al resultado final deseado

```
malla %>%
 mutate(fecha = as.Date(fecha, format = "%d/%m/%y"))
```

fecha	valor
2019-11-12	12.1
2021-01-31	13.4
2017-02-28	11.8
2018-09-03	14.4
2020-11-22	12.7

Fechas con código base

```
malla
```

fecha	valor
12-11-2019	12.1
31-01-2021	13.4
28-02-2017	11.8
03-09-2018	14.4
22-11-2020	12.7

%d-%m-%Y

```
malla %>%
  mutate(fecha = as.Date(fecha, format = "%d-%m-%Y"))
```

fecha	valor
2019-11-12	12.1
2021-01-31	13.4
2017-02-28	11.8
2018-09-03	14.4
2020-11-22	12.7

Code	Value
%d	Day of the month (decimal number)
%m	Month (decimal number)
%b	Month (abbreviated)
%B	Month (full name)
%y	Year (2 digit)
%Y	Year (4 digit)



Haga la prueba con la columna date de la malla contam

https://www.stat.berkeley.edu/~s133/dates.html



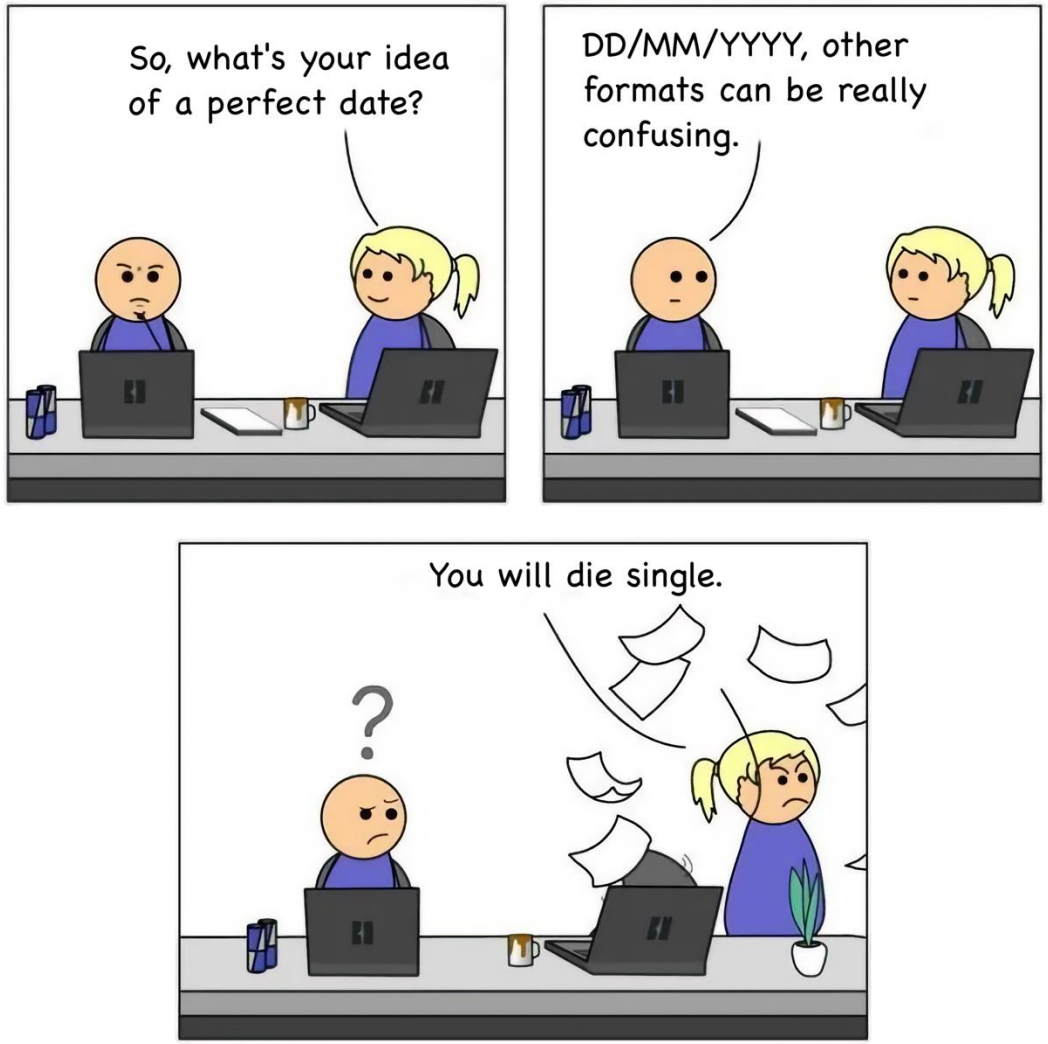

```
contam %>%
  mutate(fecha = as.Date(date, format = "%d/%m/%Y"))
```

```
# A tibble: 6,692 × 5
   date       concentration site  pollutant fecha
   <chr>              <dbl> <fct> <fct>     <date>
 1 01/01/2021            70 ACO   PM10      2021-01-01
 2 01/01/2021            44 ATI   PM10      2021-01-01
 3 01/01/2021            73 CUT   PM10      2021-01-01
 4 01/01/2021            43 FAC   PM10      2021-01-01
 5 01/01/2021            56 MER   PM10      2021-01-01
 6 01/01/2021            40 MER   PM2.5     2021-01-01
 7 01/01/2021            29 NEZ   PM2.5     2021-01-01
 8 01/01/2021            34 PED   PM10      2021-01-01
 9 01/01/2021            25 PED   PM2.5     2021-01-01
10 01/01/2021            59 SAG   PM10      2021-01-01
# ℹ 6,682 more rows
# ℹ Use `print(n = ...)` to see more rows
```

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{lubridate}

Gestión de datos con formato de fecha (date)



```
malla
```

fecha	valor
12-11-2019	12.1
31-01-2021	13.4
28-02-2017	11.8
03-09-2018	14.4
22-11-2020	12.7

```
malla %>%
  mutate(fecha = dmy(fecha))
```

fecha	valor
2019-11-12	12.1
2021-01-31	13.4
2017-02-28	11.8
2018-09-03	14.4
2020-11-22	12.7

fecha	valor
12-2019-11	12.1
31-2021-01	13.4
28-2017-02	11.8
03-2018-09	14.4
22-2020-11	12.7

```
malla %>%
  mutate(fecha = dym(fecha))
```



Haga la prueba con la columna date de la malla contam




```r
contam %>%
  mutate(fecha = dmy(date))
```

```
# A tibble: 6,692 × 5
   date       concentration site  pollutant fecha
   <chr>              <dbl> <fct> <fct>     <date>
 1 01/01/2021            70 ACO   PM10      2021-01-01
 2 01/01/2021            44 ATI   PM10      2021-01-01
 3 01/01/2021            73 CUT   PM10      2021-01-01
 4 01/01/2021            43 FAC   PM10      2021-01-01
 5 01/01/2021            56 MER   PM10      2021-01-01
 6 01/01/2021            40 MER   PM2.5     2021-01-01
 7 01/01/2021            29 NEZ   PM2.5     2021-01-01
 8 01/01/2021            34 PED   PM10      2021-01-01
 9 01/01/2021            25 PED   PM2.5     2021-01-01
10 01/01/2021            59 SAG   PM10      2021-01-01
# ℹ 6,682 more rows
# ℹ Use `print(n = ...)` to see more rows
```

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```r
today()                          # devuelve la fecha actual

year("2020-11-05")               # devuelve el año de la fecha indicada

month("2020-11-05")              # devuelve el mes de la fecha indicada

day("2020-11-05")                # devuelve el día de la fecha indicada

wday("2020-11-05")               # devuelve el número de la semana de la fecha indicada

yday("2020-11-05")               # devuelve el número de día del año de la fecha indicada
```

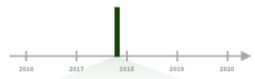
```r
malla %>%
  mutate(mes = month(fecha, label = TRUE))
```

fecha	valor	mes
2019-11-12	12.1	nov
2021-01-31	13.4	ene
2017-02-28	11.8	feb

Dates and times with lubridate : : CHEAT SHEET

https://lubridate.tidyverse.org

Date-times



`2017-11-28 12:00:00`

2017-11-28 12:00:00
A **date-time** is a point on the timeline, stored as the number of seconds since 1970-01-01 00:00:00 UTC

dt <- ***as_datetime****(1511870400)*
"2017-11-28 12:00:00 UTC"

2017-11-28
A **date** is a day stored as the number of days since 1970-01-01

d <- ***as_date****(17498)*
"2017-11-28"

12:00:00
An hms is a **time** stored as the number of seconds since 00:00:00

*t <- hms::****as.hms****(85)*
00:01:25

PARSE DATE-TIMES (Convert strings or numbers to date-times)

1. Identify the order of the year (**y**), month (**m**), day (**d**), hour (**h**), minute (**m**) and second (**s**) elements in your data.
2. Use the function below whose name replicates the order. Each accepts a wide variety of input formats.

`2017-11-28T14:02:00` **ymd_hms**(), **ymd_hm**(), **ymd_h**(). *ymd_hms("2017-11-28T14:02:00")*

`2017-22-12 10:00:00` **ydm_hms**(), **ydm_hm**(), **ydm_h**(). *ydm_hms("2017-22-12 10:00:00")*

`11/28/2017 1:02:03` **mdy_hms**(), **mdy_hm**(), **mdy_h**(). *mdy_hms("11/28/2017 1:02:03")*

`1 Jan 2017 23:59:59` **dmy_hms**(), **dmy_hm**(), **dmy_h**(). *dmy_hms("1 Jan 2017 23:59:59")*

`20170131` **ymd**(), **ydm**(). *ymd(20170131)*

`July 4th, 2000` **mdy**(), **myd**(). *mdy("July 4th, 2000")*

`4th of July '99` **dmy**(), **dym**(). *dmy("4th of July '99")*

`2001: Q3` **yq**() Q for quarter. *yq("2001: Q3")*

`2:01` hms::**hms**() Also lubridate::**hms**(), **hm**() and **ms**(), which return periods.* *hms::hms(sec = 0, min= 1, hours = 2)*

`2017.5` **date_decimal**(decimal, tz = "UTC") *date_decimal(2017.5)*

now(tzone = "") Current time in tz (defaults to system tz). *now()*

today(tzone = "") Current date in a tz (defaults to system tz). *today()*

fast_strptime() Faster strptime. *fast_strptime('9/1/01', '%y/%m/%d')*

parse_date_time() Easier strptime. *parse_date_time("9/1/01", "ymd")*

GET AND SET COMPONENTS

Use an accessor function to get a component. *d ## "2017-11-28"*
day(d) ## 28

Assign into an accessor function to change a component in place. *day(d) <- 1*
d ## "2017-11-01"

`2018-01-31 11:59:59` **date**(x) Date component. *date(dt)*

`2018-01-31 11:59:59` **year**(x) Year. *year(dt)*
isoyear(x) The ISO 8601 year.
epiyear(x) Epidemiological year.

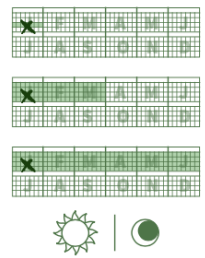
`2018-01-31 11:59:59` **month**(x, label, abbr) Month. *month(dt)*

`2018-01-31 11:59:59` **day**(x) Day of month. *day(dt)*
wday(x,label,abbr) Day of week.
qday(x) Day of quarter.

`2018-01-31 11:59:59` **hour**(x) Hour. *hour(dt)*

`2018-01-31 11:59:59` **minute**(x) Minutes. *minute(dt)*

`2018-01-31 11:59:59` **second**(x) Seconds. *second(dt)*



week(x) Week of the year. *week(dt)*
isoweek() ISO 8601 week.
epiweek() Epidemiological week.

quarter(x, with_year = FALSE) Quarter. *quarter(dt)*

semester(x, with_year = FALSE) Semester. *semester(dt)*

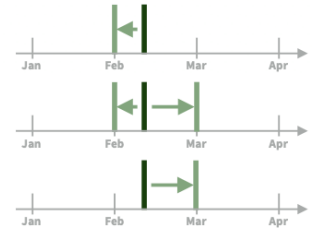
am(x) Is it in the am? *am(dt)*
pm(x) Is it in the pm? *pm(dt)*

dst(x) Is it daylight savings? *dst(d)*

leap_year(x) Is it a leap year? *leap_year(d)*

update(object, ..., simple = FALSE) *update(dt, mday = 2, hour = 1)*

Round Date-times



floor_date(x, unit = "second") Round down to nearest unit. *floor_date(dt, unit = "month")*

round_date(x, unit = "second") Round to nearest unit. *round_date(dt, unit = "month")*

ceiling_date(x, unit = "second", change_on_boundary = NULL) Round up to nearest unit. *ceiling_date(dt, unit = "month")*

rollback(dates, roll_to_first = FALSE, preserve_hms = TRUE) Roll back to last day of previous month. *rollback(dt)*

Stamp Date-times

stamp() Derive a template from an example string and return a new function that will apply the template to date-times. Also **stamp_date**() and **stamp_time**().

1. Derive a template, create a function
   sf <- stamp("Created Sunday, Jan 17, 1999 3:34")
2. Apply the template to dates
   sf(ymd("2010-04-05"))
   ## [1] "Created Monday, Apr 05, 2010 00:00"

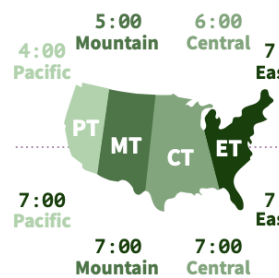
Tip: use a date with day > 12

Time Zones

R recognizes ~600 time zones. Each encodes the time zone, Daylight Savings Time, and historical calendar variations for an area. R assigns *one* time zone per vector.

Use the **UTC** time zone to avoid Daylight Savings.

OlsonNames() Returns a list of valid time zone names. *OlsonNames()*



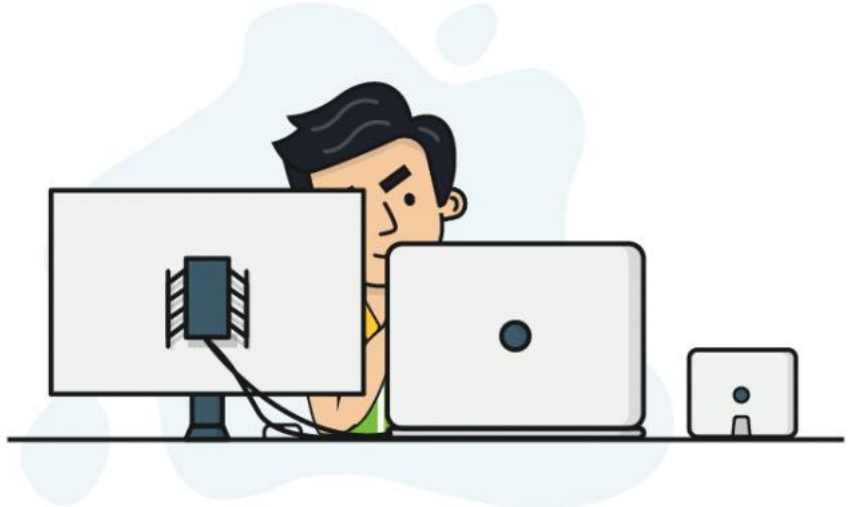
with_tz(time, tzone = "") Get the **same date-time** in a new time zone (a new clock time). *with_tz(dt, "US/Pacific")*

force_tz(time, tzone = "") Get the **same clock time** in a new time zone (a new date-time). *force_tz(dt, "US/Pacific")*

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Su turno...

- Convierta a formato de fecha la columna date
- Genere columnas day, month (con etiqueta), year
- Guarde el objeto con el mismo nombre
- Genere el summary de la malla de datos



```r
contam <-
  contam %>%
  mutate(
    date = dmy(date),
    day = day(date),
    month = month(date, label = TRUE),
    year = year(date)
  ) %>%
  print()
```

```
# A tibble: 6,692 × 7
   date       concentration site  pollutant   day month  year
   <date>             <dbl> <fct> <fct>     <int> <ord> <dbl>
 1 2021-01-01            70 ACO   PM10          1 Jan    2021
 2 2021-01-01            44 ATI   PM10          1 Jan    2021
 3 2021-01-01            73 CUT   PM10          1 Jan    2021
 4 2021-01-01            43 FAC   PM10          1 Jan    2021
 5 2021-01-01            56 MER   PM10          1 Jan    2021
 6 2021-01-01            40 MER   PM2.5         1 Jan    2021
 7 2021-01-01            29 NEZ   PM2.5         1 Jan    2021
 8 2021-01-01            34 PED   PM10          1 Jan    2021
 9 2021-01-01            25 PED   PM2.5         1 Jan    2021
10 2021-01-01            59 SAG   PM10          1 Jan    2021
# ℹ 6,682 more rows
# ℹ Use `print(n = ...)` to see more rows
```

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Malla incial

```
summary(contam)
```

```
       date          id_station      id_parameter      value              unit
 Length   :6692   Length   :6692   Length   :6692   Min.   :  2.00   Min.   :2
 N.unique : 243   N.unique :  28   N.unique :   2   1st Qu.: 17.00   1st Qu.:2
 N.blank  :   0   N.blank  :   0   N.blank  :   0   Median : 25.00   Median :2
 Min.nchar:  10   Min.nchar:   3   Min.nchar:   4   Mean   : 30.46   Mean   :2
 Max.nchar:  10   Max.nchar:   3   Max.nchar:   5   3rd Qu.: 39.00   3rd Qu.:2
                                                    Max.   :158.00   Max.   :2
```

Malla procesada

```
summary(contam)
```

```
     date            concentration          site      pollutant         day              month          year
Min.   :2021-01-01   Min.   :  2.00   BJU     : 482   PM10 :3529   Min.   : 1.00   Jul     : 919   Min.   :2021
1st Qu.:2021-03-06   1st Qu.: 17.00   MER     : 478   PM2.5:3163   1st Qu.: 8.00   May     : 917   1st Qu.:2021
Median :2021-05-07   Median : 25.00   PED     : 470                Median :16.00   Jun     : 851   Median :2021
Mean   :2021-05-04   Mean   : 30.46   TLA     : 432                Mean   :15.69   Mar     : 843   Mean   :2021
3rd Qu.:2021-07-03   3rd Qu.: 39.00   SFE     : 386                3rd Qu.:23.00   Jan     : 834   3rd Qu.:2021
Max.   :2021-08-31   Max.   :158.00   CAM     : 364                Max.   :31.00   Aug     : 831   Max.   :2021
                                      (Other):4080                                 (Other):1497
```

Su turno...

- Genere una nueva malla de datos que mantenga sólo los datos de $PM_{2.5}$
- Renombre la columna concentration como pm25
- Preserve sólo las columnas date, site y pm25
- Nombre y guarde al objeto como pm25_2021
- Exporte su malla pm25_2021 con el nombre pm25_cdmx_2021 con extensión .rds y .csv



```r
pm25_2021 <-
  contam %>%
  filter(pollutant == "PM2.5") %>%
  rename(pm25 = concentration) %>%
  select(date, site, pm25) %>%
  print()
```

```
# A tibble: 3,163 × 3
   date       site   pm25
   <date>     <fct> <dbl>
 1 2021-01-01 MER      40
 2 2021-01-01 NEZ      29
 3 2021-01-01 PED      25
 4 2021-01-01 SAG      36
 5 2021-01-01 SFE      13
 6 2021-01-01 TLA      27
 7 2021-01-01 UAX      33
 8 2021-01-01 UIZ      40
 9 2021-01-01 CCA      26
10 2021-01-01 BJU      30
# ℹ 3,153 more rows
# ℹ Use `print(n = ...)` to
```

```r
write_rds(pm25_2021, "./data/pm25_cdmx_2021.rds")
write_excel_csv(pm25_2021, "./data/pm25_cdmx_2021.csv")
```